



the operating system of the machines, causing these to fraudulently dispense currency on the attacker's command.

During the remote attack, malware was installed on the target system via exploited vulnerabilities in the remote management system, most notably the use of default passwords and remote management transmission control protocol (TCP) ports. The attacker then executed the malware, causing the target ATM to dispense a given amount of currency.

Jack has also developed software that allows him to remotely send an electric shock to anyone wearing a pacemaker within a 15 metres radius.

Jack has also come up with a system that scans for insulin pumps that communicate wirelessly within a 91 metres radius, allows hackers to hack into these without needing to know the identification numbers and then sets these to dish out more or less insulin than necessary, sending patients into hypoglycaemic shock or ketoacidosis.

With more such hackers who can find out the vulnerabilities in various medical devices, hacking the human body is becoming pretty easy.

World's first digital weapon

Stuxnet, a highly sophisticated computer worm was discovered in 2010 and was essentially termed the world's first digital weapon. It was developed by American and Israeli governments, and was used to wreak havoc on an Iranian nuclear facility called Natanz. It targets industrial control systems that are used to monitor and control large-scale industrial facilities like power plants, dams, waste-processing systems and similar operations. It allows attackers to take control of these systems without the operators knowing.

Attacking Stuxnet was the first one that allowed hackers to manipulate real-world equipment, thereby making it very dangerous. It was the first computer virus to be able to wreak havoc in the physical world. It was sophisticated, well-funded and

there were not many groups that could pull off this kind of threat.

It was also the first cyber attack that specifically targeted industrial control systems. It targeted the computer system of machines used to enrich uranium, known as centrifuges, and instructed these to spin the machines out of control. Eventually, the forces broke the centrifuges.

At the same time, Stuxnet reported to the control room that nothing was amiss. Over a period of few years, about 20 per cent of Iran's centrifuges spun out of control and were destroyed. Iran's nuclear scientists had no idea why so many centrifuges were busted. It indeed turned out to be a brilliantly sophisticated attack.

A kindle of devices attacked at once

A massive Distributed Denial of Service (DDoS) attack against Dyn, a major domain name system (DNS) provider, broke large portions of the Internet in October 2016, causing significant outage to a ton of websites and services, including Twitter, GitHub, PayPal, Amazon, Reddit, Netflix and Spotify. According to security intelligence firm Flashpoint, Mirai bots were detected driving much, but not necessarily all, of the traffic in the DDoS attacks against DynDNS.

Mirai is a piece of malware that targets IoT devices such as routers, security cameras and DVRs, and enslaves vast numbers of these compromised devices into a botnet, which is then used to conduct DDoS attacks. Since the source code of Mirai botnet has already been made available to the public, anyone can wield these attacks against targets.

This time hackers did not target an individual site, rather, they attacked Dyn that many sites and services were using as their upstream DNS provider for turning IP addresses into human-readable websites.

This type of attack is notable and concerning because it largely consists of unsecured IoT devices,

which are growing exponentially with time. These devices are implemented in a way that these cannot easily be updated and, thus, are nearly impossible to secure.

Manufacturers majorly focus on performance and usability of IoT devices but ignore security measures and encryption mechanisms, which is why these are routinely being hacked and widely becoming part of DDoS botnets used as weapons in cyber attacks.

An online tracker of Mirai botnet suggests that there are more than 1.2 million Mirai-infected devices on the Internet, with over 166,000 devices active right now.

Not-so-smart refrigerator

White-hat hackers, or ethical computer hackers, at Pen-Test Partners were able to use fake security credentials to intercept communications between the fridge and Google Calendar.

Besides the fridge, the hackers also found 25 vulnerabilities in 14 allegedly smartdevices including scales, coffee makers, wireless cameras, locks, home automation hubs and fingerprint readers.

The hack was pulled off against RF28HMEBSR smartfridge, part of Samsung's line-up of Smart Home appliances that can be controlled via their Smart Home app. While the fridge implements SSL, it fails to validate SSL certificates; thereby, enabling man-in-the-middle attacks against most connections.

The Internet-connected device is designed to download Gmail Calendar information to an onscreen display. Security shortcomings mean that hackers who manage to jump on to the same network can potentially steal Google login credentials from their neighbours.

To sum it, the threats are real and our data is prone to attacks. Therefore while using IoT devices make sure that security is built on the foundation of the system and validity checks, authentication, data verification and encryption is carried out frequently. **EFY**